Class5: Data Viz with ggplot2

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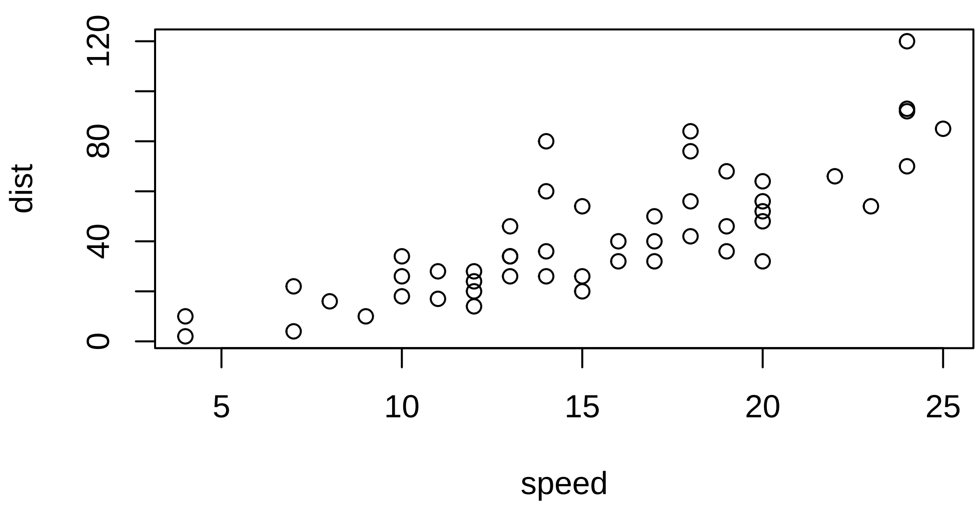
#Background

There are many graphics systems in R for making plots and figures. These include so-called *"base R" graphics* like the `plot()` function and add on packages like **gglplot2**.

Let's compare how we make a simple figure with these two systems:

We can use the in-built 'cars' dataset:

```
plot(cars)
```



Before I can use ggplot 2 I need to install it on my computer.

To do this we can use the function `install.packages("ggplot2")`

>**N.B.** We never run `install.packages()`in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our report.

Once installed we need to load up the package into our R brain:

```r
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

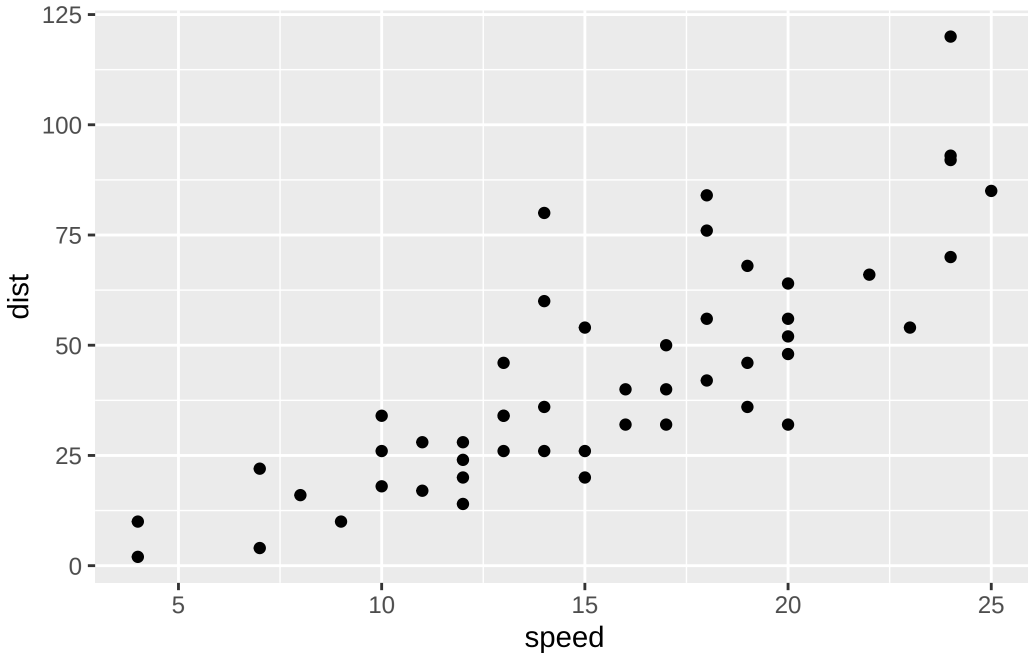
The main function in the ggplot2 package is called `ggplot()`

```r
ggplot(cars)
```


Every ggplot has at least 3 layers:

- The **data** (a data.frame of the stuff we want to plot)
- The **aes**thetics (how the data maps to the plot)
- The **geom layer (how you want the plot drawn, e.g. points, lines, etc.)**

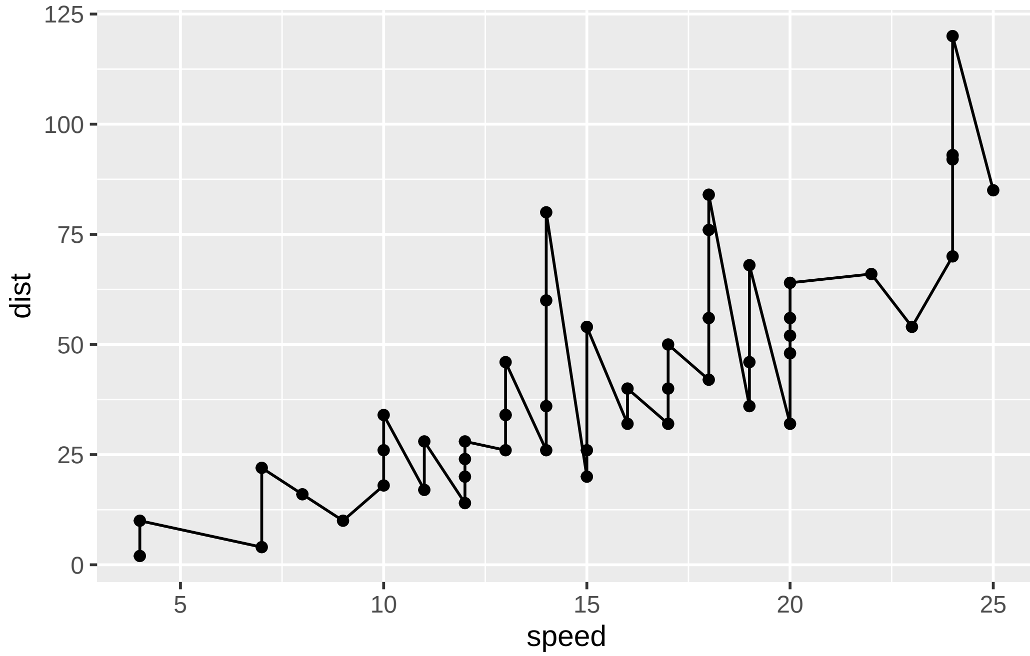
```r
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



Add some custom features

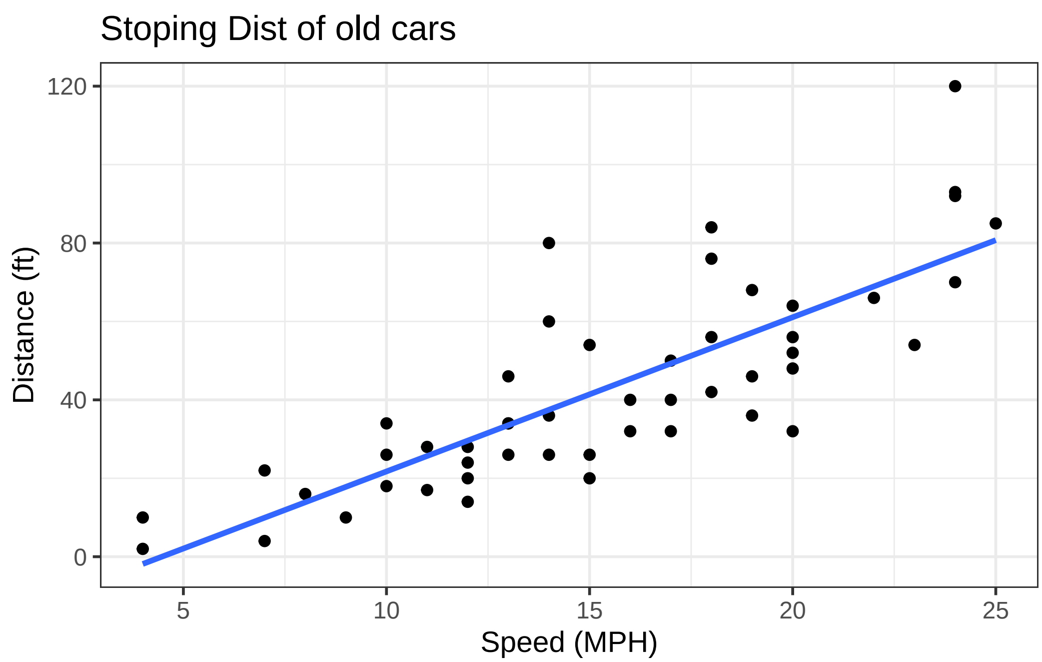
Let's add a trend line that shows the relationship between speed and distance.

```r
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_line()
```




```r
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw() +
  labs(title="Stoping Dist of old cars",
       x="Speed (MPH)",
       y="Distance (ft)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Q. Can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the "gray" error region.

Important data

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
sum(genes$State == "up")
```

```
[1] 127
```

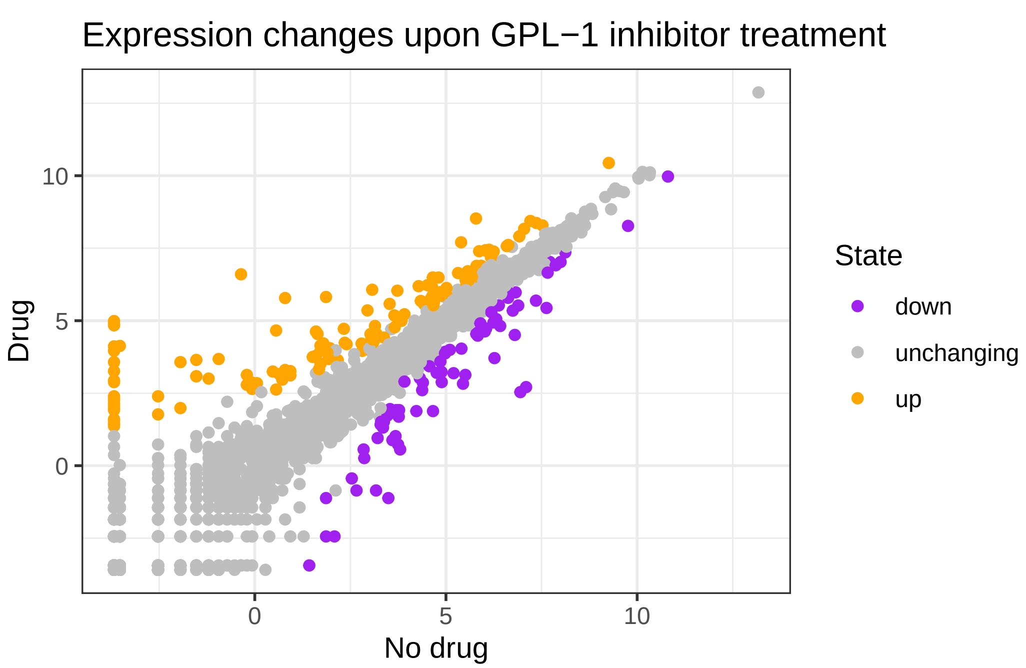
A useful new function in this context is the `table()` function:

```r
table(genes$State)
```

```
      down unchanging         up 
        72       4997        127 
```

My first plot attempt

```r
ggplot(genes) + 
  aes(Condition1, Condition2, col=State) + 
  geom_point() + 
  scale_color_manual(values=c("purple", "gray", "orange")) + 
  theme_bw()+
  labs(x="No drug", y="Drug",
       title="Expression changes upon GPL-1 inhibitor treatment")
```



Going further

Here we read the famous gapminder data set:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries (i.e. rows) are in this data set?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different “country” entries are in this dataset?

```
length(unique(gapminder$country))
```

```
[1] 142
```

```
unique(gapminder$country)
```

```
  [1] "Afghanistan"              "Albania"
  [3] "Algeria"                  "Angola"
  [5] "Argentina"                "Australia"
  [7] "Austria"                  "Bahrain"
  [9] "Bangladesh"               "Belgium"
 [11] "Benin"                    "Bolivia"
 [13] "Bosnia and Herzegovina"   "Botswana"
 [15] "Brazil"                   "Bulgaria"
 [17] "Burkina Faso"             "Burundi"
 [19] "Cambodia"                 "Cameroon"
 [21] "Canada"                   "Central African Republic"
 [23] "Chad"                     "Chile"
```

```
 [25] "China"                    "Colombia"
 [27] "Comoros"                  "Congo, Dem. Rep."
 [29] "Congo, Rep."              "Costa Rica"
 [31] "Cote d'Ivoire"            "Croatia"
 [33] "Cuba"                     "Czech Republic"
 [35] "Denmark"                  "Djibouti"
 [37] "Dominican Republic"       "Ecuador"
 [39] "Egypt"                    "El Salvador"
 [41] "Equatorial Guinea"        "Eritrea"
 [43] "Ethiopia"                 "Finland"
 [45] "France"                   "Gabon"
 [47] "Gambia"                   "Germany"
 [49] "Ghana"                    "Greece"
 [51] "Guatemala"                "Guinea"
 [53] "Guinea-Bissau"            "Haiti"
 [55] "Honduras"                 "Hong Kong, China"
 [57] "Hungary"                  "Iceland"
 [59] "India"                    "Indonesia"
 [61] "Iran"                     "Iraq"
 [63] "Ireland"                  "Israel"
 [65] "Italy"                    "Jamaica"
 [67] "Japan"                    "Jordan"
 [69] "Kenya"                    "Korea, Dem. Rep."
 [71] "Korea, Rep."              "Kuwait"
 [73] "Lebanon"                  "Lesotho"
 [75] "Liberia"                  "Libya"
 [77] "Madagascar"               "Malawi"
 [79] "Malaysia"                 "Mali"
 [81] "Mauritania"               "Mauritius"
 [83] "Mexico"                   "Mongolia"
 [85] "Montenegro"               "Morocco"
 [87] "Mozambique"               "Myanmar"
 [89] "Namibia"                  "Nepal"
 [91] "Netherlands"              "New Zealand"
 [93] "Nicaragua"                "Niger"
 [95] "Nigeria"                  "Norway"
 [97] "Oman"                     "Pakistan"
 [99] "Panama"                   "Paraguay"
[101] "Peru"                     "Philippines"
[103] "Poland"                   "Portugal"
[105] "Puerto Rico"              "Reunion"
[107] "Romania"                  "Rwanda"
[109] "Sao Tome and Principe"    "Saudi Arabia"
```

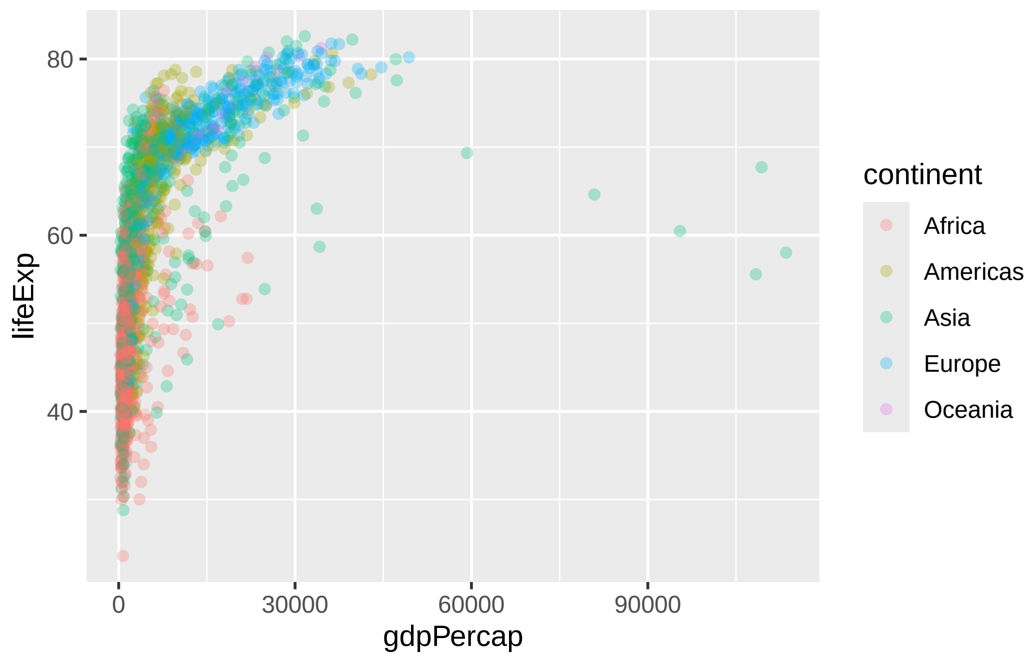
```
[111] "Senegal"                  "Serbia"
[113] "Sierra Leone"             "Singapore"
[115] "Slovak Republic"          "Slovenia"
[117] "Somalia"                  "South Africa"
[119] "Spain"                    "Sri Lanka"
[121] "Sudan"                    "Swaziland"
[123] "Sweden"                   "Switzerland"
[125] "Syria"                    "Taiwan"
[127] "Tanzania"                 "Thailand"
[129] "Togo"                     "Trinidad and Tobago"
[131] "Tunisia"                  "Turkey"
[133] "Uganda"                   "United Kingdom"
[135] "United States"            "Uruguay"
[137] "Venezuela"                "Vietnam"
[139] "West Bank and Gaza"       "Yemen, Rep."
[141] "Zambia"                   "Zimbabwe"
```

Let's make our first plot of the entire dataset:

Plot of "gdpPercap" vs "lifeExp" colored by "continent"

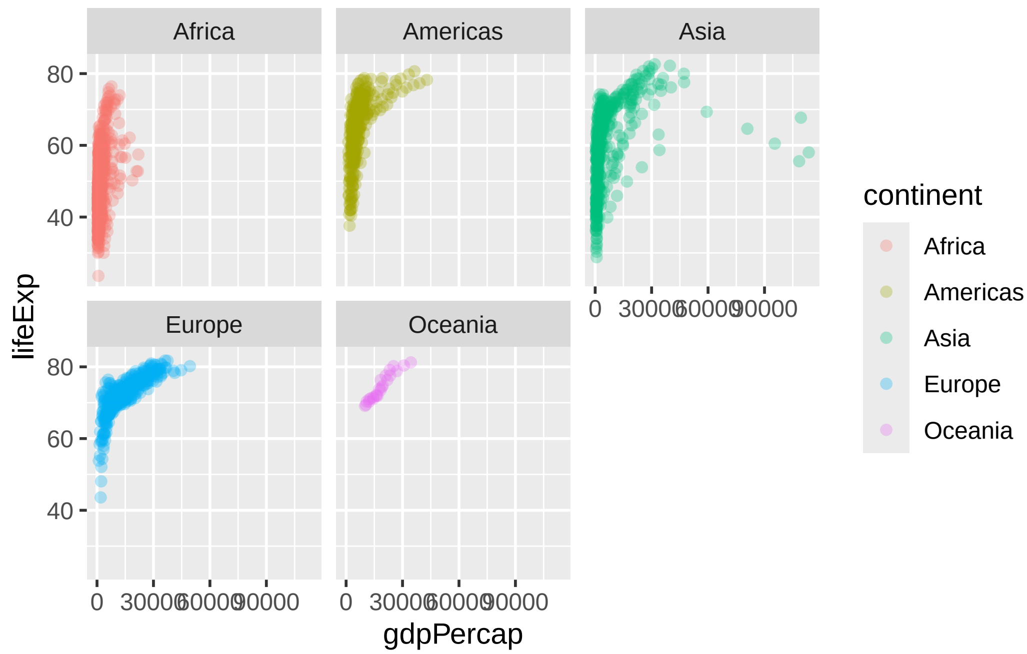
```r
p <- ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
  geom_point(alpha=0.3)
```

```r
p
```



I can add more layers to ‘p‘

```
p +
  facet_wrap(~continent)
```



Make a plot for 1977 and 2007 only (not all the years in the data set).

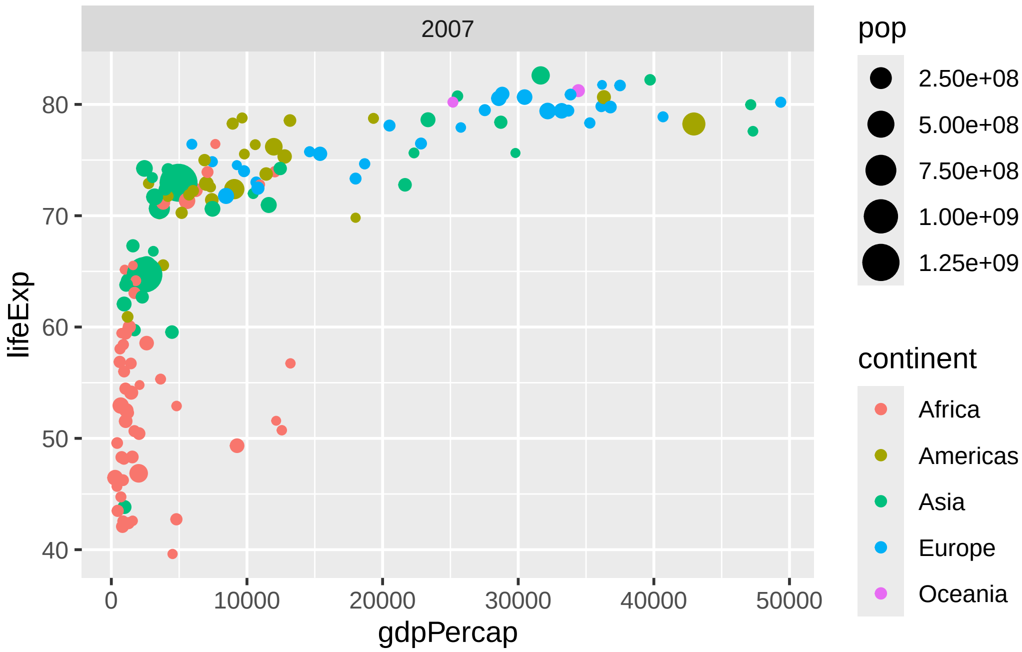
> Q. First use the **dplyr** package and the `filter()` function from that package to extract the year 2007.

```
library (dplyr)
```

```
Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag


The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
g07 <- filter(gapminder, year ==2007)
g77 <- filter(gapminder, year ==1977)
g <- filter(gapminder, year ==2007 | year ==1977)
```

```
ggplot(g07) +
  aes(gdpPercap, lifeExp, col=continent, size=pop) +
  geom_point() +
  facet_wrap(~year)
```

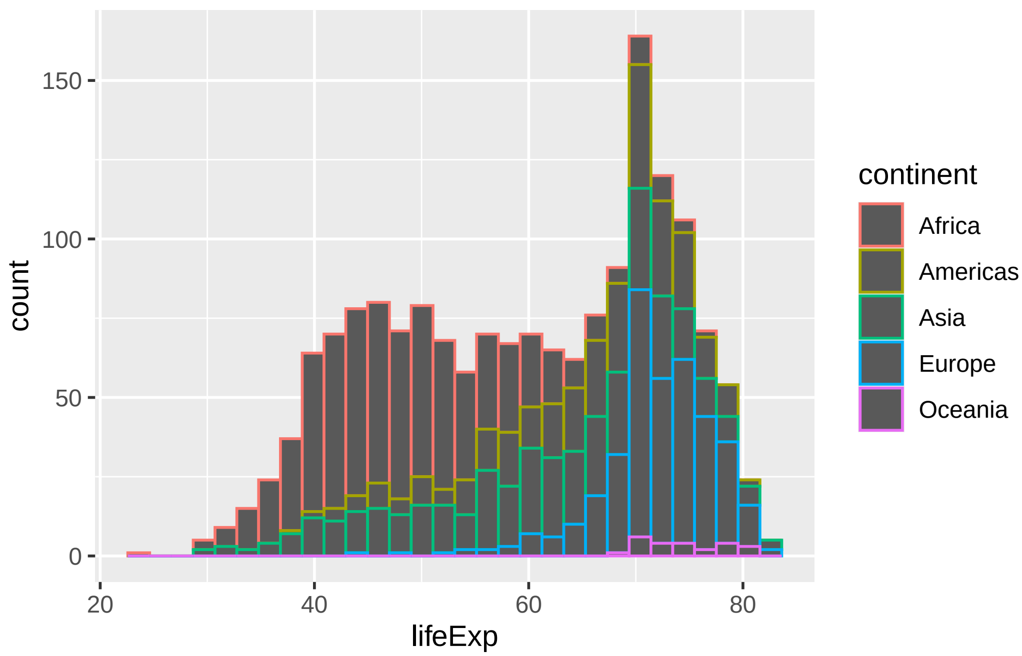


Q, Make a histogram of lifeExp colored by continent (try using `fill=continent` or `col=continent`)

Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
  aes(x = lifeExp, col = continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```


count
150
100
50
0
20
40
60
80
lifeExp
continent
Africa
Americas
Asia
Europe
Oceania